

distribution comes with over 250 packages automatically installed, with the option of installing over 7500 additional open source packages. The most important thing to remember is that whatever package or library you have installed by using Anaconda can be used in your Jupyter Notebook also. Whatever updation is required by other packages or libraries during the installation of a new package will be automatically handled by Anaconda carefully, without bothering you in the least.

So, finally, we have reached a place in our journey where we no longer need to worry about installing the new packages and libraries necessary to continue our quest for developing AI and machine learning based applications. Notice that Anaconda only has a command-line interface (CLI). Are we in trouble? No! Our installation also gives us Anaconda Navigator, a graphical user interface (GUI) for Anaconda. Execute the command '*anaconda-navigator*' on the terminal to run Anaconda Navigator (Figure 1). Soon, we will see an example of its power.

Introduction to Theano

Now, let us try to work with Theano, a Python library and optimising compiler for evaluating mathematical expressions. The installation of Theano is very easy since we have Anaconda Navigator. First, open Anaconda Navigator. On the top right corner you will see the button *Environments* (marked with a red box in Figure 1). Press that button. You will go to a window showing the list of all currently installed packages. From the drop-down list on top, choose the option *Not installed*. Scroll down, find the option *Theano*, and click on the checkbox on the left. Now press the green button named *Apply* at the bottom right corner of the window. Anaconda will find out all the dependencies for installing Theano and show them in a pop-up menu. Figure 2 shows the pop-up menu I got during the installation of Theano in my system. You can see that in addition to Theano, one more new package is installed and eight other packages are modified. Imagine how difficult this would have been if I was trying to

manually install Theano. But with the help of Anaconda, all we need to do is press the button *Apply* on the bottom right corner of this pop-up menu. After a while, the installation of Theano will be completed smoothly. Now, we are ready to use Theano in our Jupyter Notebooks.

We are already familiar with the Python library called *SymPy* used for symbolic calculation but Theano takes it to the next level. Let us see a simple example to understand the power of Theano. Consider the code shown in Figure 3. First, let us try to understand the code. Line 1 of the code imports Theano. Line 2 imports *tensor* and names it *T*. We have already encountered tensors when we discussed TensorFlow. Mathematically, tensors can be treated as multi-dimensional arrays. *Tensor* is one of the key data structures used in Theano and can be used to store and manipulate scalars (numbers), vectors (1-dimensional arrays), matrices (2-dimensional arrays), tensors (multi-dimensional arrays), etc. In Line 3, a Theano function called *function()* is imported. Line 4 imports a Theano function called *pp()*, which is used for pretty-printing (printing in a format appealing to humans). Line 5 creates a symbolic scalar variable of type double called *x*. It is a bit tricky to understand these sorts of symbolic variables. Think of it as a 0-sized double variable with no value or storage associated. Similarly, Line 6 creates another symbolic scalar variable called *y*. Line 7 acts like a function in a sense. This line tells the Python interpreter something like the following, 'if and when symbolic scalar variables *x* and *y* get some values, add those values and store inside me'.

To further illustrate symbolic operations at this point, let us try to understand Line 8. From Figure 3, it is clear that the output of this line of code is $(x+y)$. So, it is clear that actual addition of two numbers is yet to take place. Lines 9 to 11 similarly define symbolic subtraction, multiplication,

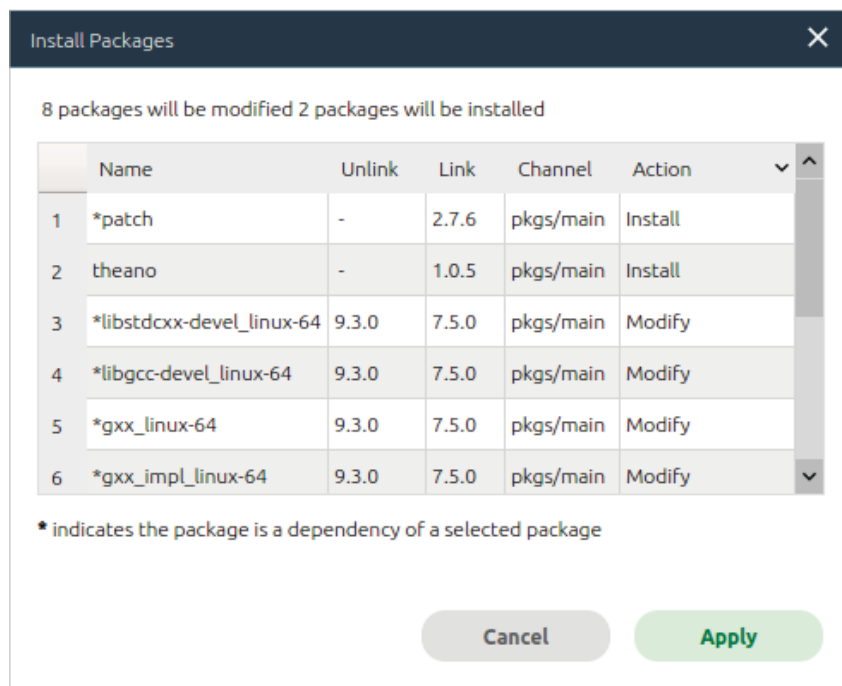


Figure 2: Installation of Theano